722. CLINICAL ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION: POSTER I | NOVEMBER 29, 2018

Intestinal Microbiota at Neutrophil Engraftment Can Predict Acute Graft-Versus-Host Disease in the Recipients of Allogeneic Hematopoietic Stem Cell Transplantation

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Abstract

Backgrounds: An increasing of researches have demonstrated that intestinal microbiome plays an important role in the development of acute graft-versus-host disease (aGVHD), and loss of microbiota diversity is associated with aGVHD. But whether microbiota might be as a biomarker to predict aGVHD is undetermined. This study investigates the prediction of intestinal microbiota at neutrophil engraftment for occurrence of aGVHD.

Methods: We prospectively collected stool and blood samples from 166 patients undergoing allogeneic hematopoietic stem cell transplantation (allo-HSCT) at pre-conditioning, day 0 and day 15 post-transplantation. The microbiota in stool were detected by 16S rRNA gene sequencing; the inflammatory factors (TNF-α, IL-6, IL-17A, IL-1β and Lipopolysaccharide (LPS)) in blood by multiplex immunoassays and dynamic turbidimetric methods; and T lymphocyte subsets were tested by flow cytometry. AGVHD was determined by retrospective review of the clinical charts.

Results: 141 patients were enrolled into analysis, including 83 in grade 0-I aGVHD (non-aGVHD group) and 58 in grades II-IV aGVHD (aGVHD group). The microbiota diversity was lower in aGVHD group than that in non-aGVHD both at day 0 and day 15 post-transplantation (P = 0.018, 0.009, respectively). The diversity was associated with conditioning (Intensified vs. Standard: medain, 3.39 vs. 3.04 (P = 0.017, day 0); medain, 3.17 vs.1.89 ($P = 0.045$, day 15)) and β-lactam antibiotics (β-lactam vs. No β-lactam: medain, 3.33 vs. 2.48, $P = 0.004$ (day 15)). At day 15 post-transplantation, Treg/Th17 cells ratio in blood positively correlated with the microbial diversity($r = 0.217$, $P = 0.010$), but negatively correlated with the levels of TNF-α, IL-17A and LPS ($r = -0.226, -0.216, -0.291$; $P = 0.007, 0.010, < 0.001$, respectively). Based on the accumulated microbiota Score of the diversity and the special discriminated constitution (Enterobacteriaceae, Lachnospiraceae, Peptostreptococcaceae and Erysipelotrichaceae) between aGVHD and non-aGVHD groups at day 15, the area under the receiver operator characteristic curve (AUC) of the Score could be used to predict aGVHD (II-IV aGVHD: AUC = 0.75, $P < 0.001$; III-IV aGVHD: AUC = 0.84, $p < 0.001$). The Score also correlated with the grades of aGVHD ($r = 0.481$, $P < 0.001$).

Conclusions: The intestinal microbiota diversity is affected by both conditioning and antibiotics. The constitution of microbiota at neutrophil engraftment was a predictive marker for the subsequent development and grades of aGVHD.

Keywords: intestinal microbiota, marker, acute graft-versus-host disease, hematopoietic stem cell transplantation

Disclosures

No relevant conflicts of interest to declare.

Author notes

* Asterisk with author names denotes non-ASH members.